LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Sound

Chapter / Topic: Sound (topic-based)

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SECTION A

Q1. Sound is produced by: **[1]**

Answer: (b) vibrating objects

Q2. The unit of frequency is the: **[1]**

Answer: (d) hertz

Q3. The loudness of a sound depends on its: **[1]**

Answer: (a) amplitude

Q4. Sound cannot travel through: **[1]**

Answer: (a) a vacuum

Q5. In humans, sound is produced by the: **[1]**

Answer: (d) larynx (voice box)

Q6. A vibration has a time period of 0.05 s. Its frequency is: **[1]**

Answer: (c) 20 Hz

SECTION B

Q7. Why can't astronauts on the Moon talk to each other directly, without radios? **[2]**

Answer: There is no air on the Moon. Sound needs a medium to travel, so it cannot pass through the empty space between them.

- No air on the Moon: 1
- Sound needs a medium: 1

Q8. Which part of the ear vibrates when sound reaches it? Why should we never put sharp objects into our ears? **[2]**

Answer: The eardrum. A sharp object can damage the eardrum, which can affect hearing.

- Eardrum: 1
- Damage → hearing loss: 1

Q9. What is the difference between music and noise? **[2]**

Answer: Sounds that are pleasant to hear are called music. Unpleasant sounds are called noise. Even music becomes noise when it is too loud.

- Music: 1
- Noise: 1

Q10. Give two sources of noise pollution and two of its harmful effects. **[2]**

Answer: Sources (any two): vehicle horns, loudspeakers, firecrackers, factory machines, loud TV or music. Effects (any two): lack of sleep, high blood pressure, anxiety, hearing loss.

- Two sources: 1
- Two effects: 1

SECTION C

Q11. A pendulum makes 20 oscillations in 4 seconds. **[3]**

(i) $20 \div 4 = 5$ Hz [1]

(ii) $1 \div 5 = 0.2$ s [1]

(iii) The sound becomes louder. [1]

Q12. Which part vibrates to produce sound in (a) a sitar, (b) a flute, (c) a dholak? **[3]**

Answer: (a) Sitar: the stretched strings. (b) Flute: the air column inside the tube. (c) Dholak: the stretched membrane (skin).

- Each: 1

SECTION D

Q13. Answer the following: **[5]**

(i) The voice box (larynx) at the upper end of the windpipe has two vocal cords with a narrow slit between them. When air from the lungs is forced through the slit, the vocal cords vibrate and produce sound. [2]

- Larynx with vocal cords: 1
- Air makes them vibrate: 1

(ii) Men's vocal cords are longer (about 20 mm, compared with about 15 mm in women), so they vibrate at a lower frequency, giving a lower pitch. [1]

(iii) (Any two) Plant trees along the roads around the school; put up 'No horn' signs and enforce them; keep loud vehicles and factories away; use sound-absorbing materials in classrooms. [2]

- Any two: 1 + 1